

Static Timing Analysis

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Course Overview

1. STA Concepts
 - 2 lectures
2. Delay Modeling
 - 2 lectures
3. Interconnect Parasitics
 - 2 lectures
4. Delay Calculation
 - 2 lectures
5. Configuring the STA Environment
 - 3 lectures
6. Timing Checks
 - 1 lecture
7. Crosstalk and Noise
 - 2 lectures
8. Statistical Static Timing Analysis
 - 2 lectures

8. Statistical Static Timing Analysis

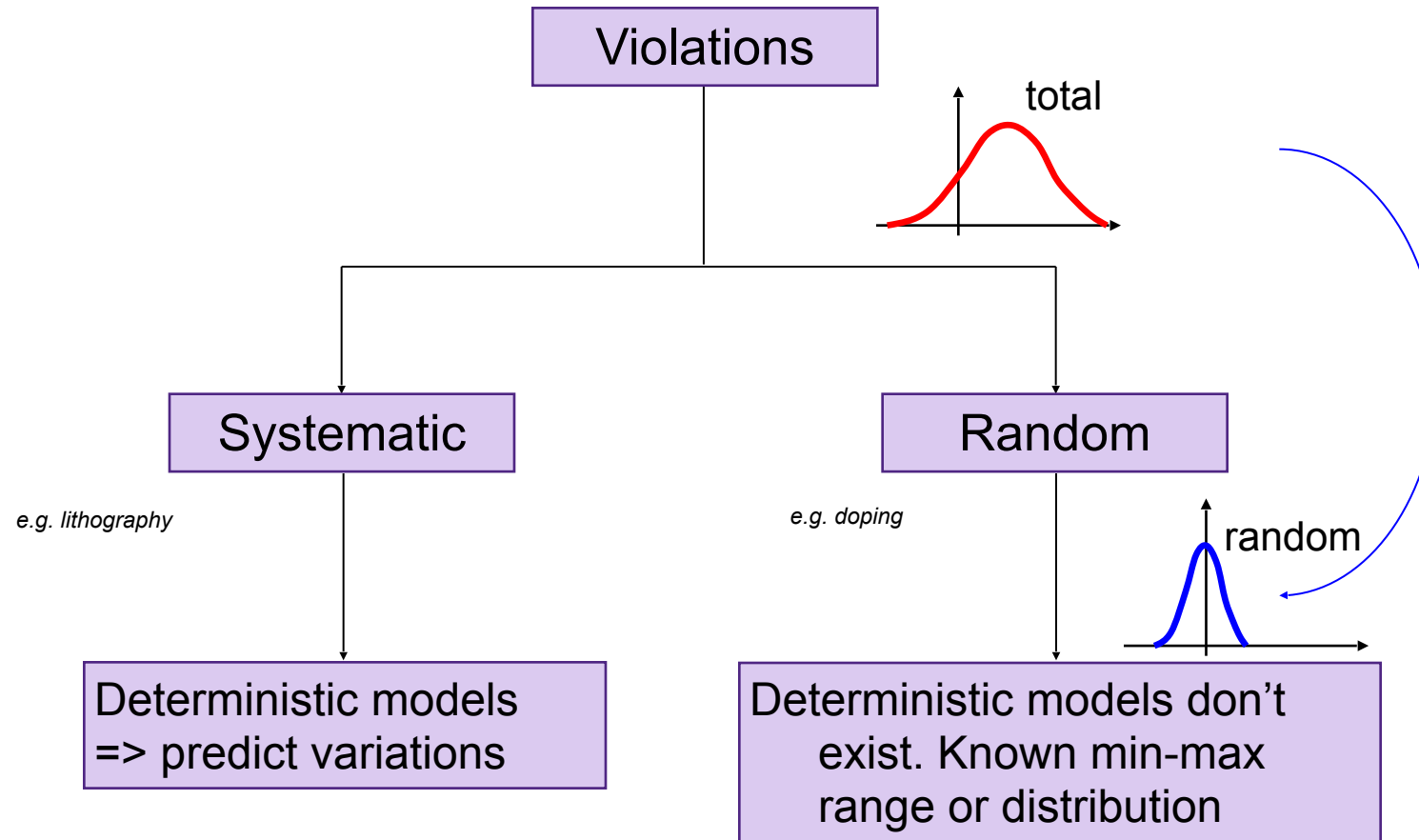
Variability

- On-chip variation (OCV)
 - Device
 - Interconnect
- Operating conditions
 - Temperature
 - Voltage
 - ...

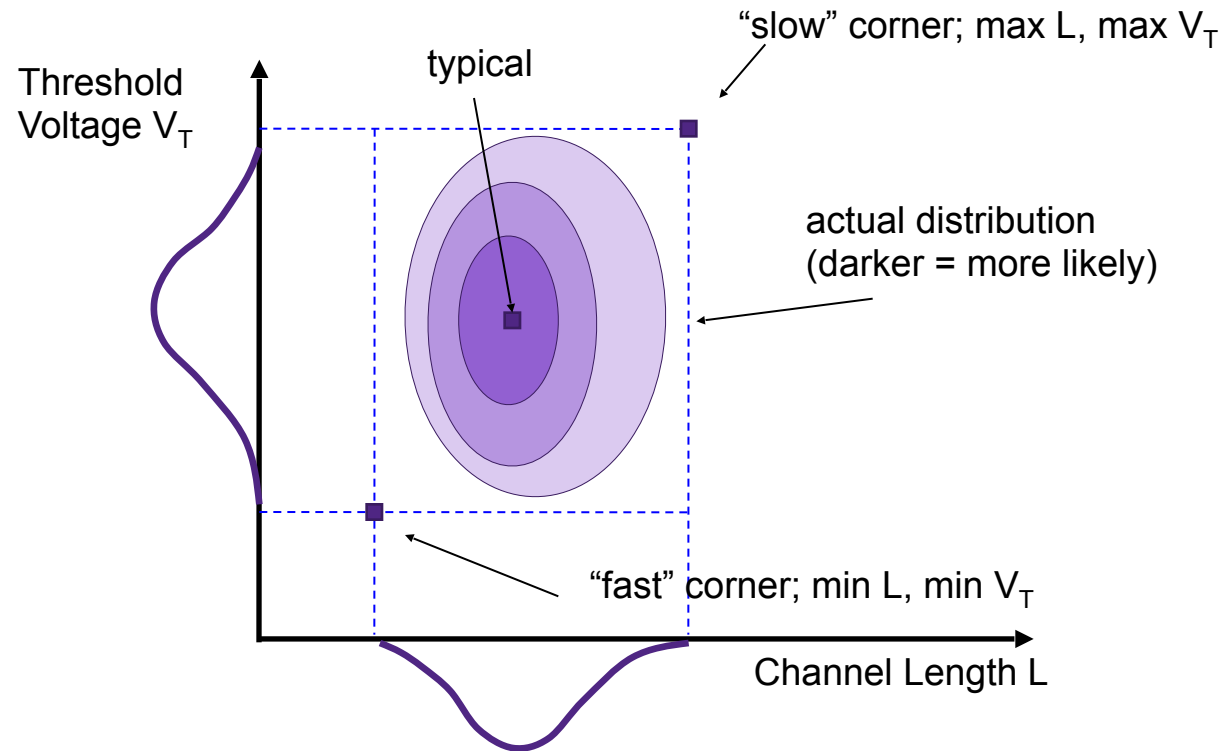
Process Variation Sources and Types

- Types
 - Systematic
 - Random
- Sources
 - Lithography induced Variations
 - Etch induced Variations
 - CMP induced Variations
 - Random Dopant Fluctuations
 - Interconnect Variations
 - Non-Linear Resistance
 - Metal Mismatch
 - ...

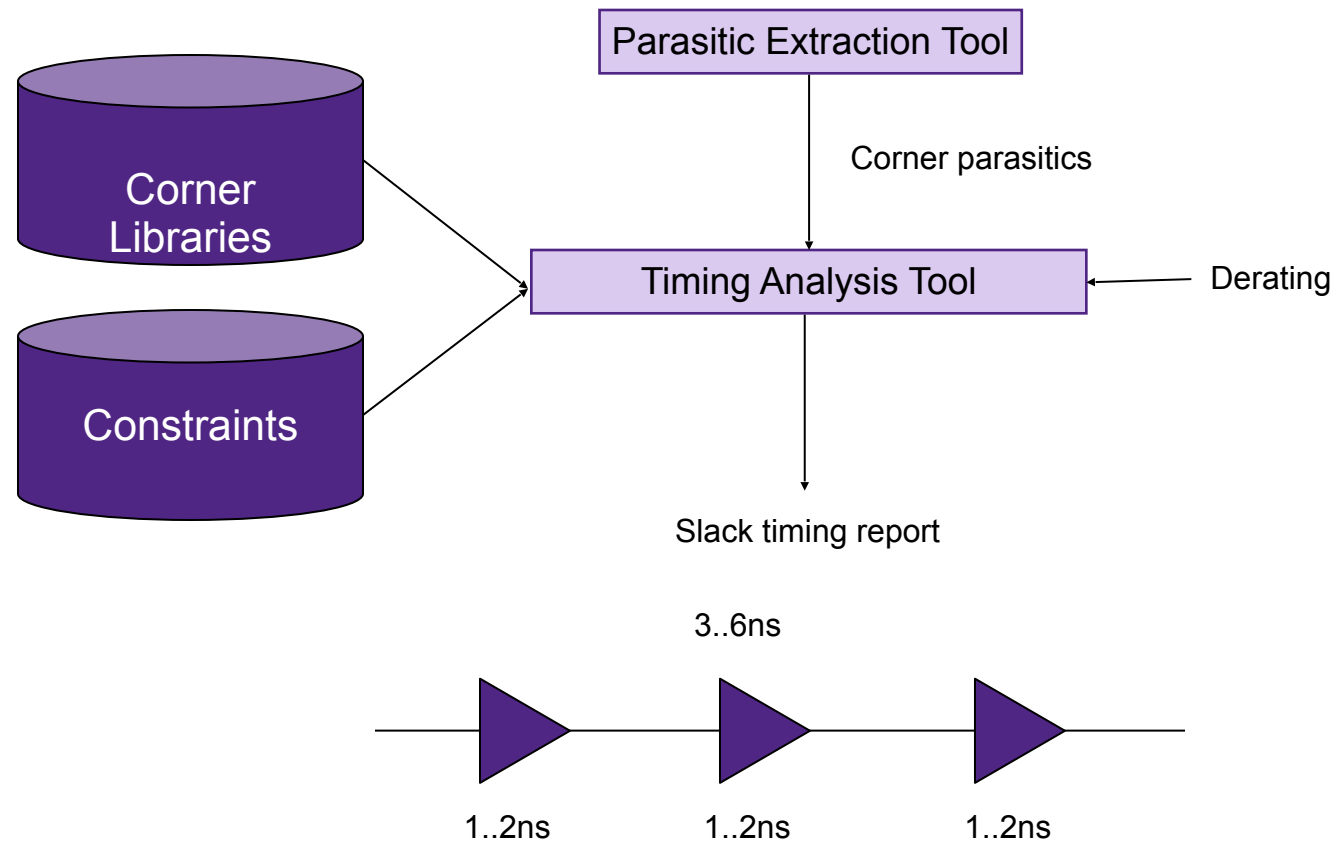
Modeling Variation



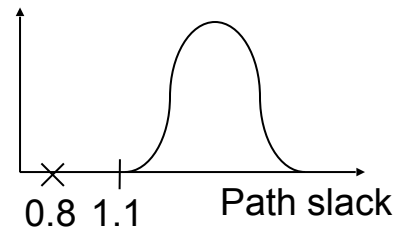
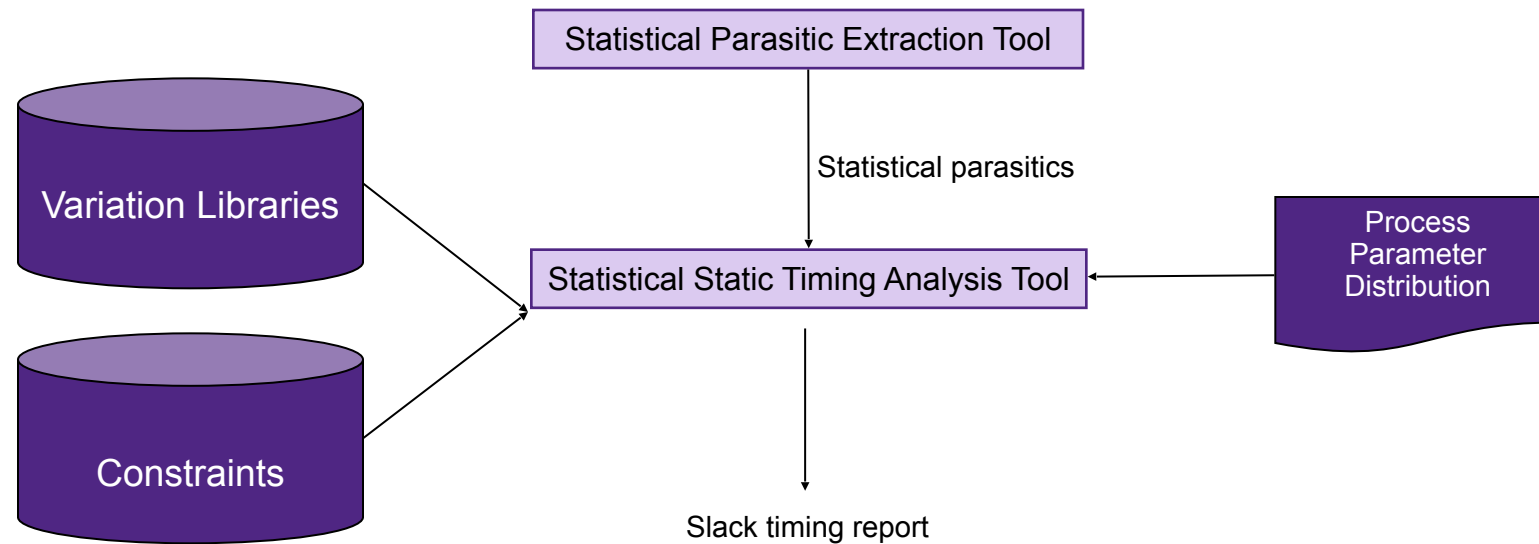
Process Corners



Conventional Corner-Based Analysis

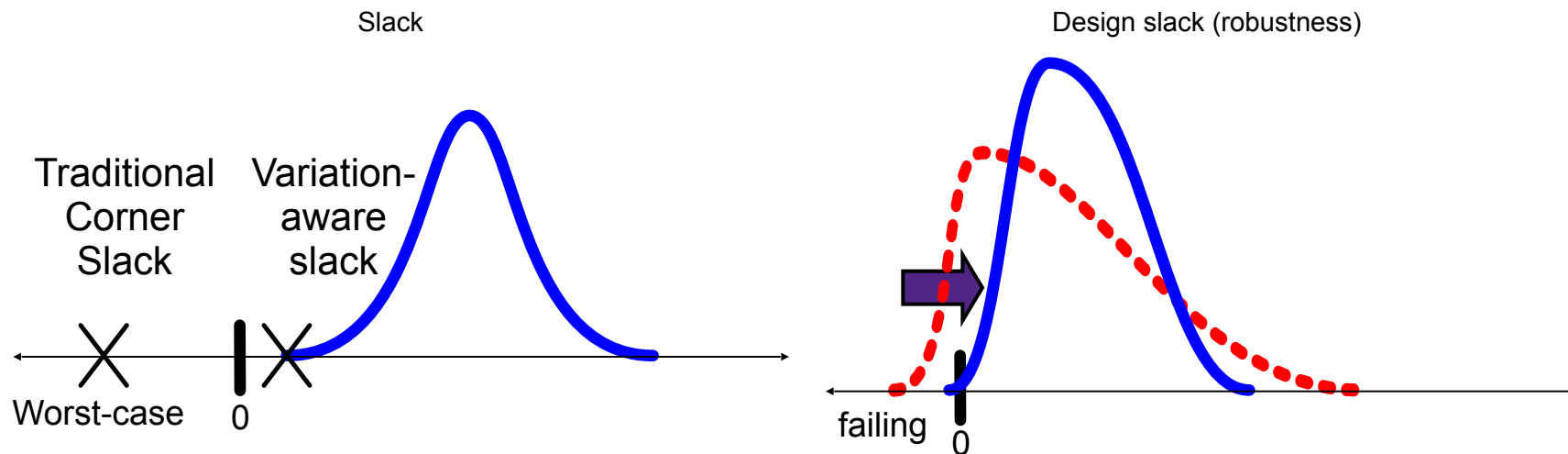
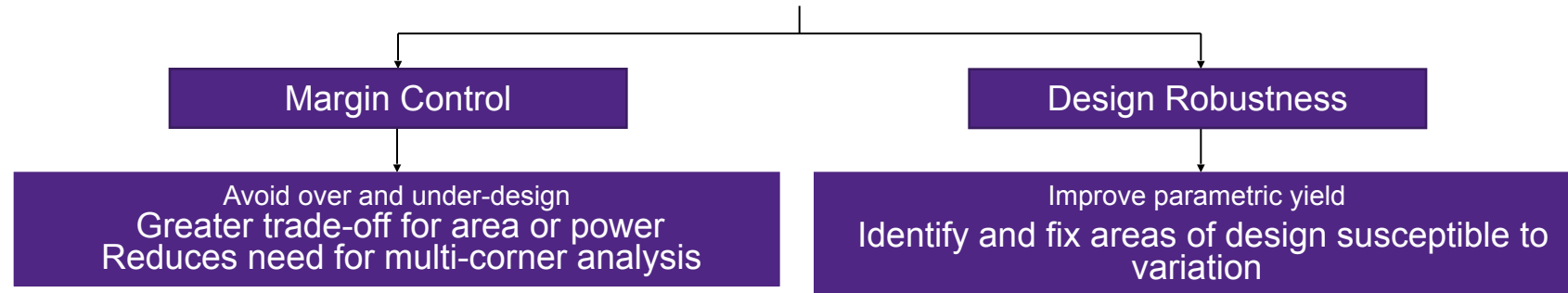


Variation-Aware Timing Analysis Flow



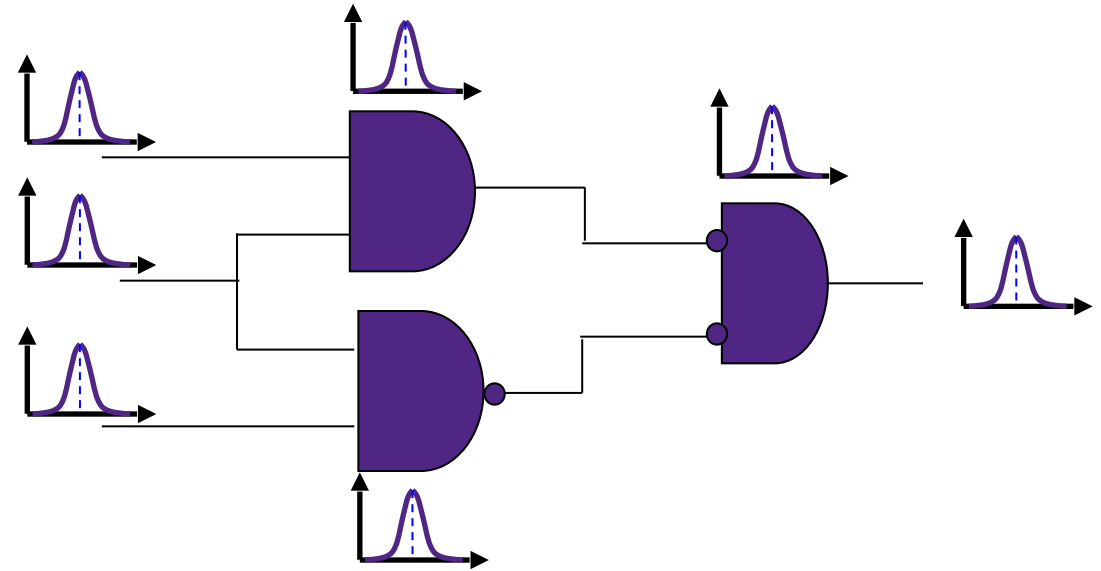
Worst case (min/max) slack: 0.8ns
97% passing slack: 1.1ns

Benefits of Variation-Aware Analysis



Everything Statistical

- Gate delays are statistical
- Signal arrival times are statistical
- Want distribution of delay at output
- Gates and signals are correlated
- Correlations are both local and global



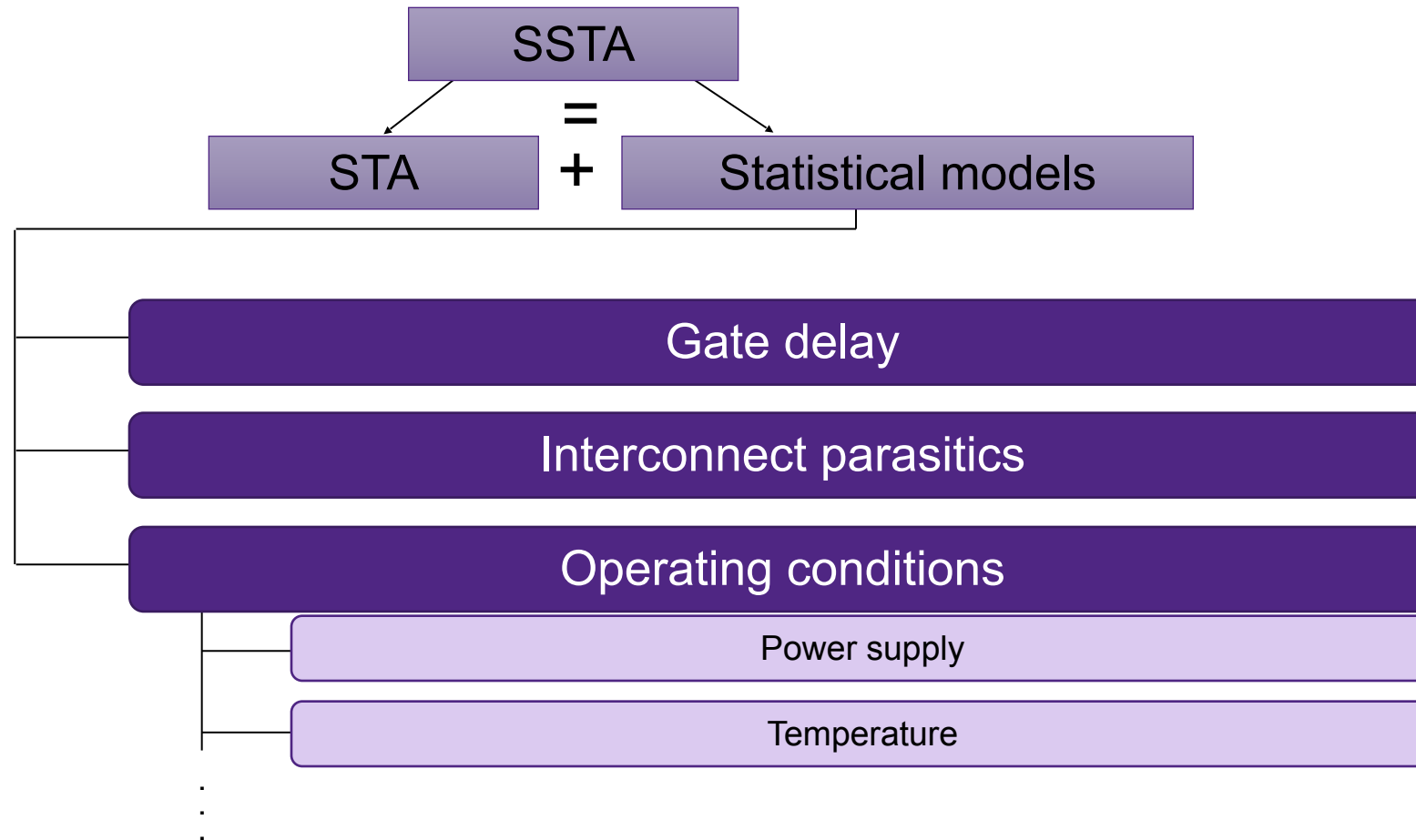
Statistical Timing Analysis

- Monte-Carlo
- Statistical Static Timing Analysis

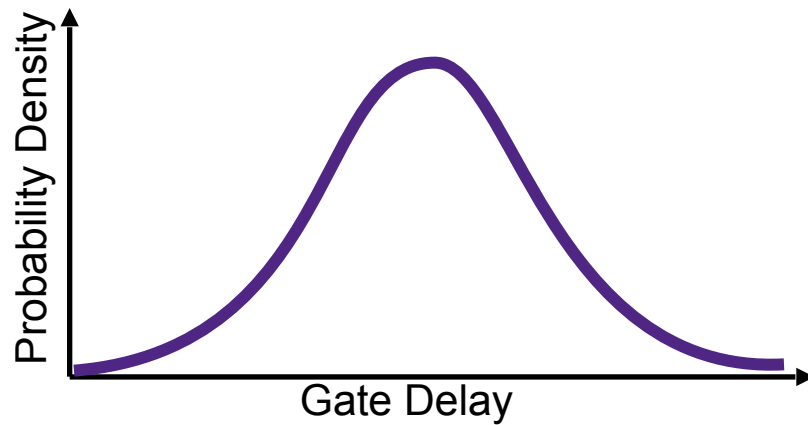
Monte-Carlo Method

- Random sampling of each probability distribution within the model
- Produces 1000's or even 10,000's of scenarios
- Iterations or trials
- Each probability distribution is sampled in a manner that reproduces the distribution's shape
- Distribution of the values calculated for the model outcome therefore reflects the probability of the values that could occur

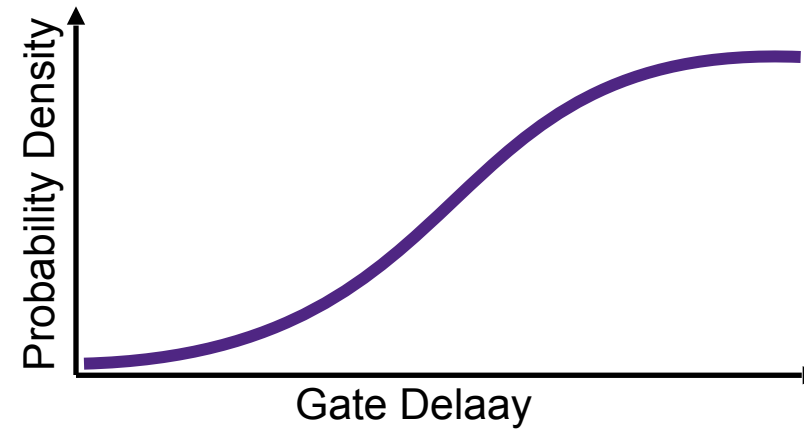
Introduction to Statistical Static Timing Analysis



Probability

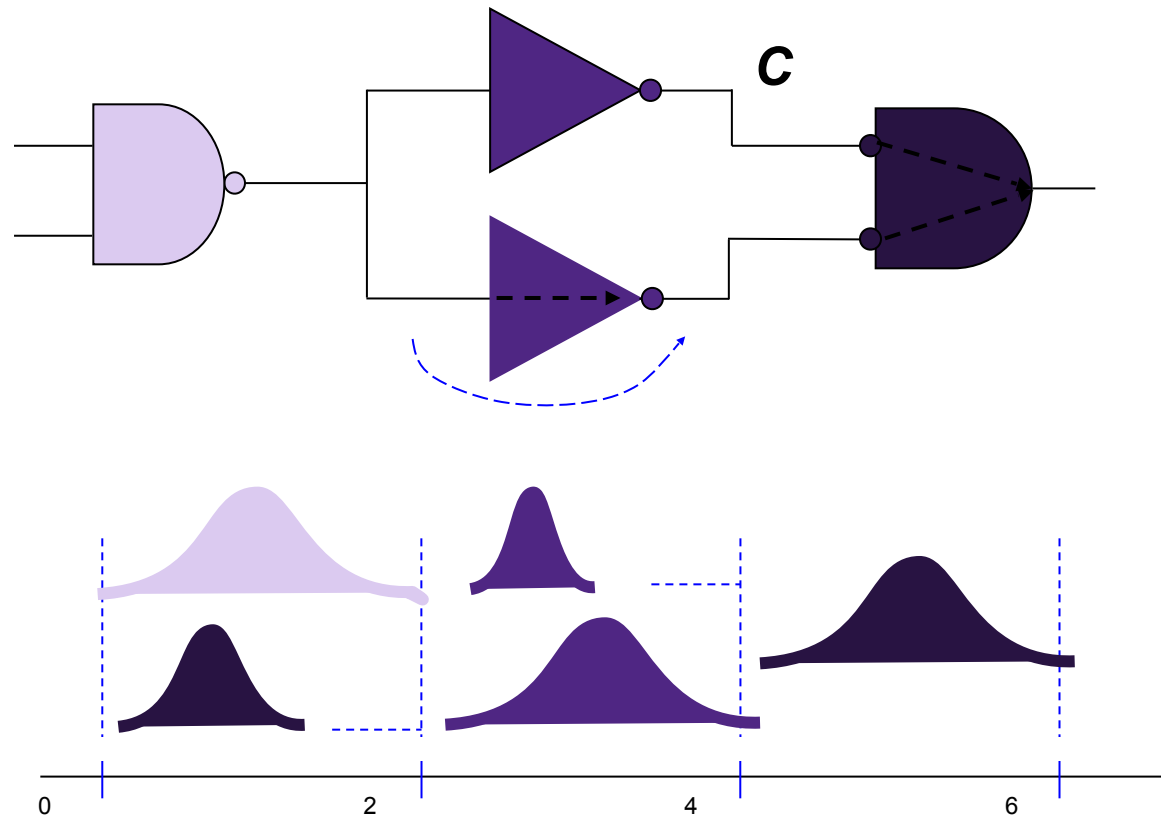


Probability Density
Function (PDF)
Probability that gate
delay has value x

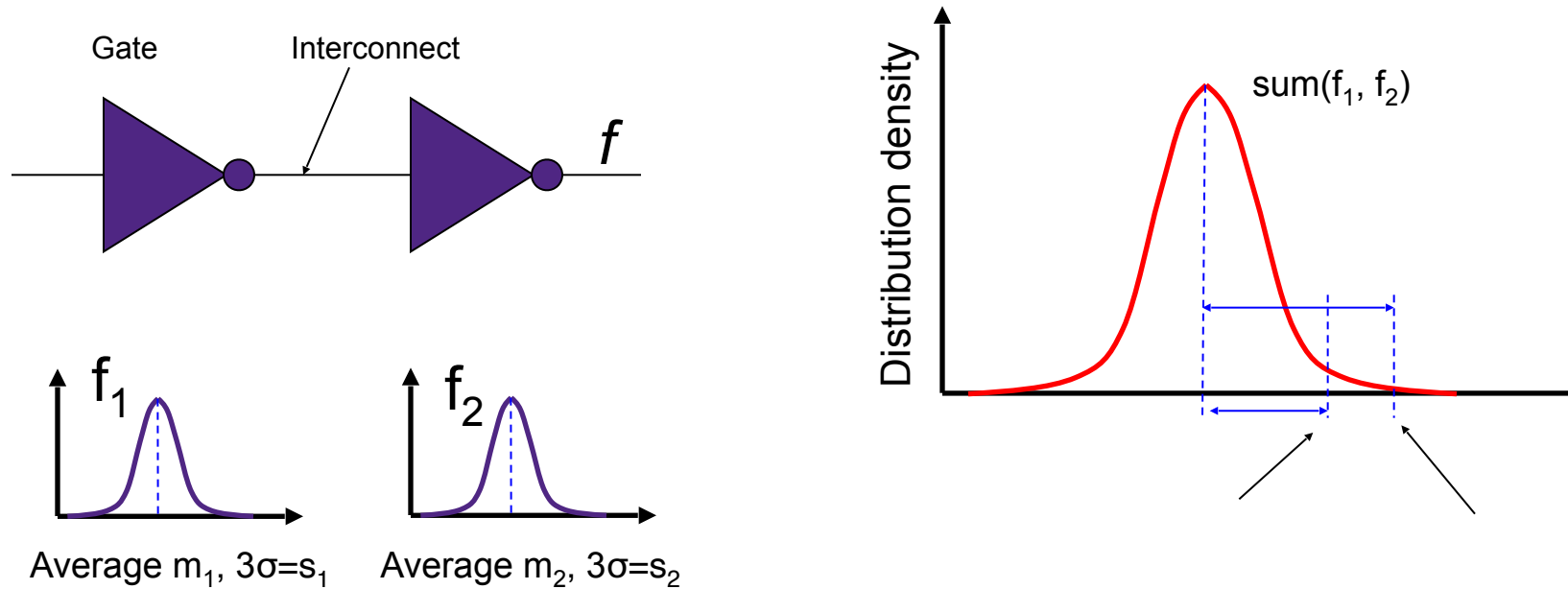


Comulative Probability
Distribution Function (PDF)
Probability that gate delay is
lower than x

Distribution of Propagation



Statistical ADD Operation

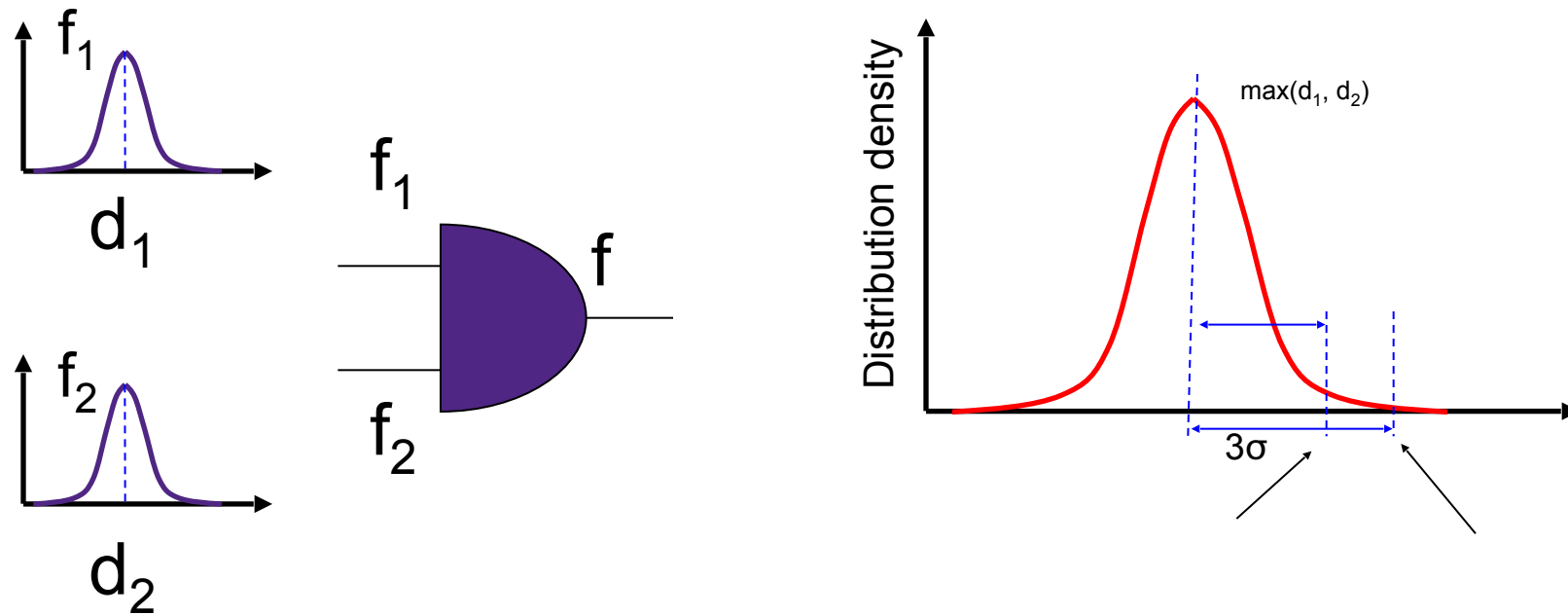


Delay PDF of path is statistical **sum** of f_1 and f_2

STA:

SSTA:

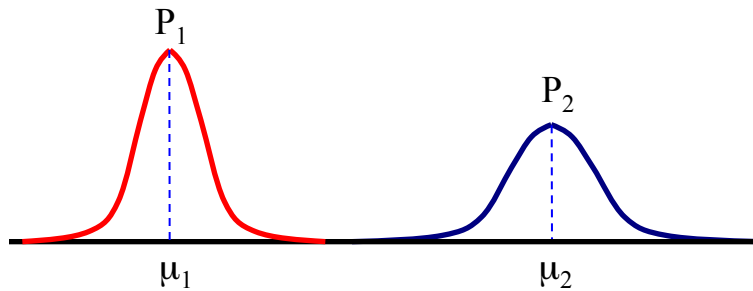
Statistical MAX Operation



Delay PDF at output pin is calculated as upper bound of statistical max of d_1 and d_2

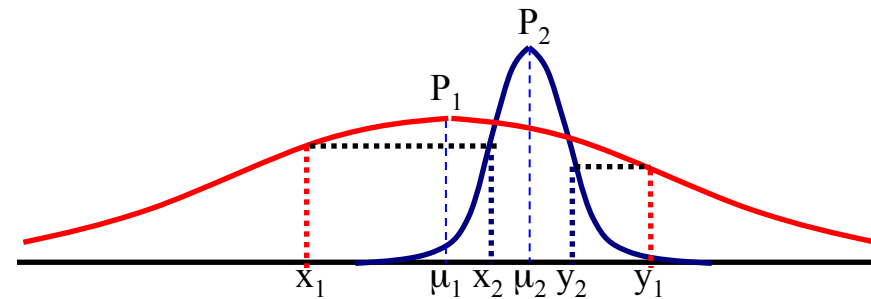
Statistical MAX Operation (2)

Non-overlapping



MAX is trivial

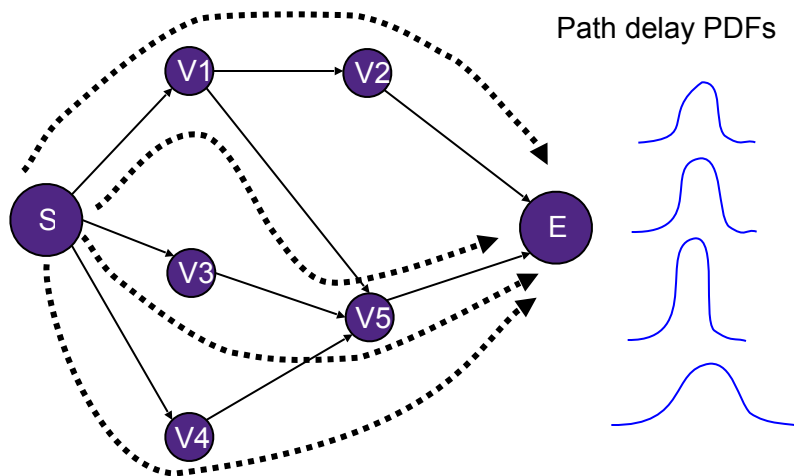
Highly correlated,
overlapping,
different sigmas



MAX is complicated

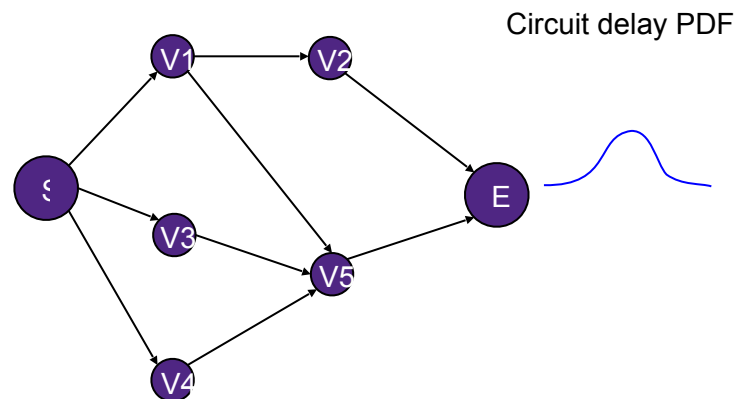
SSTA Approaches

Path-Based SSTA



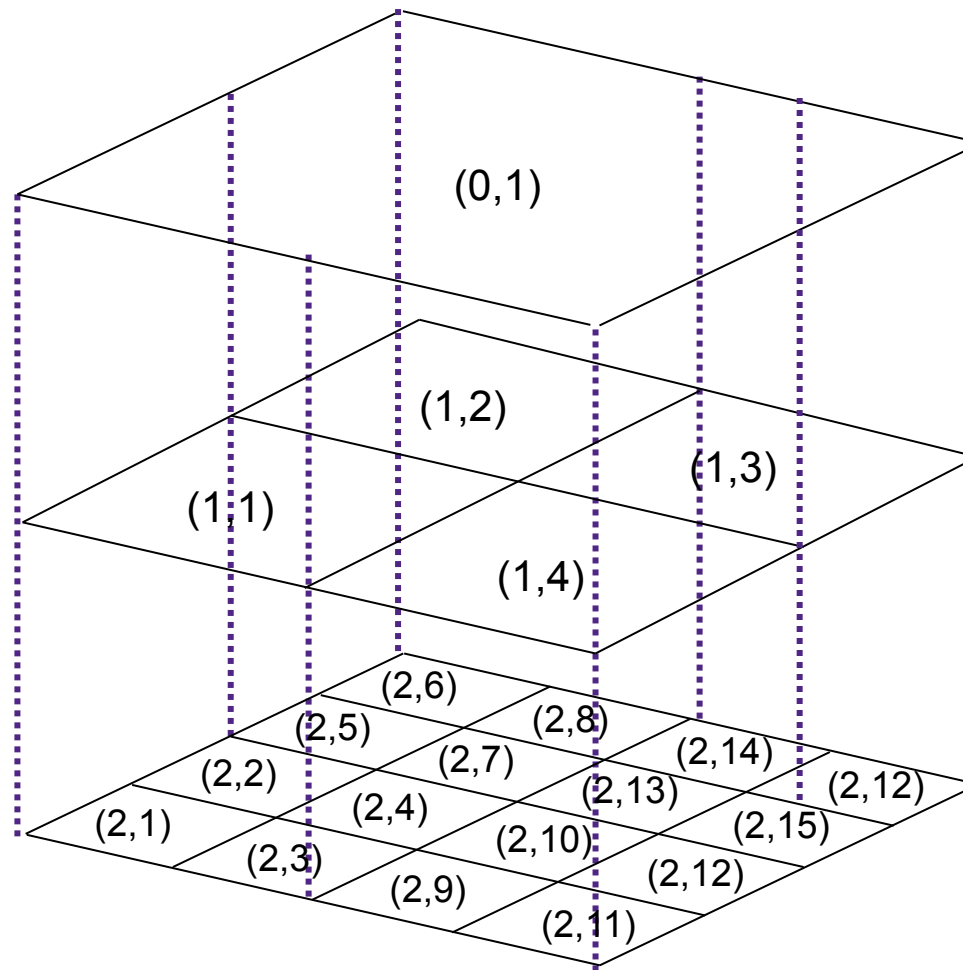
Individual analysis of paths from S to E

Block-Based SSTA

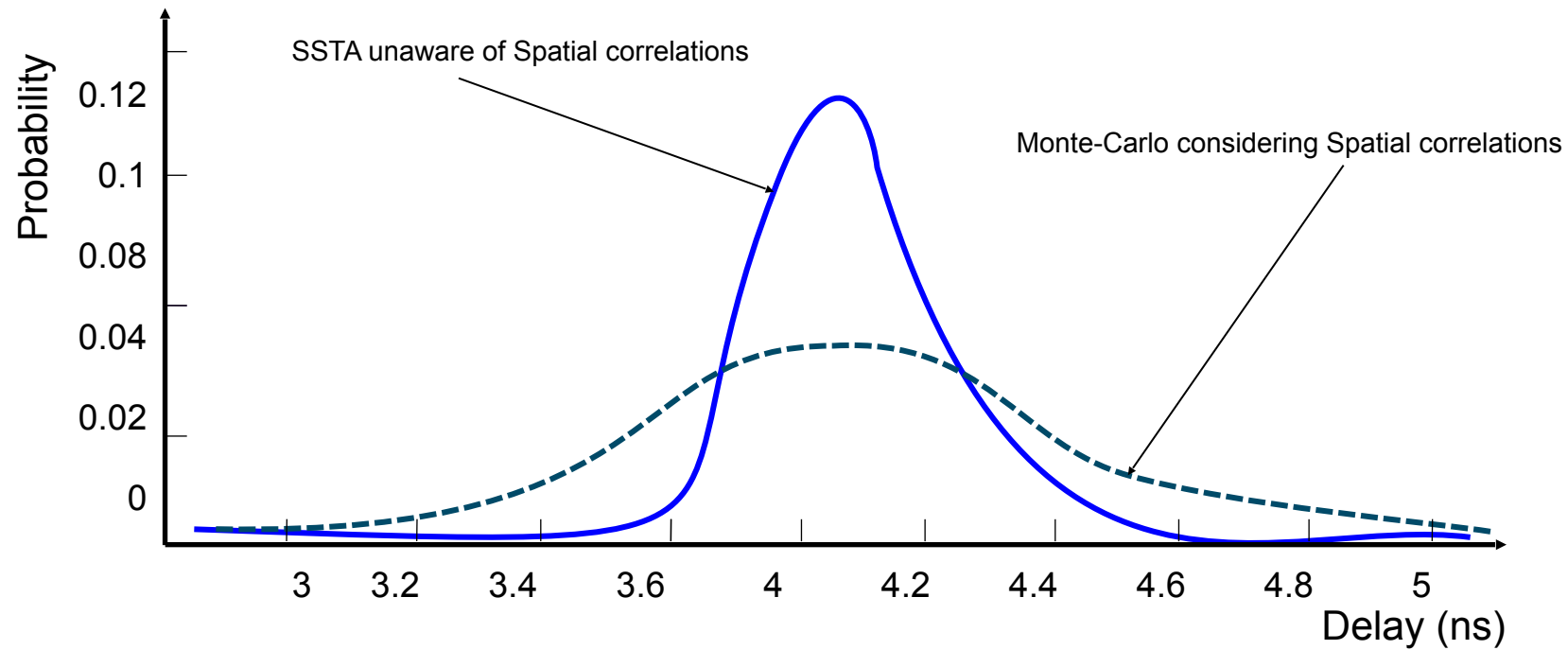


Simultaneous analysis of all paths from S to E

Spatial Correlations



Spatial Correlations Should be Considered

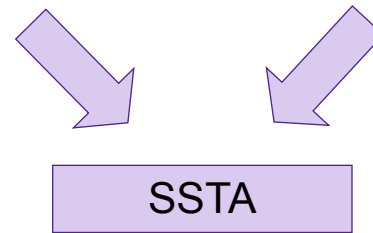


Considering Spatial Correlations

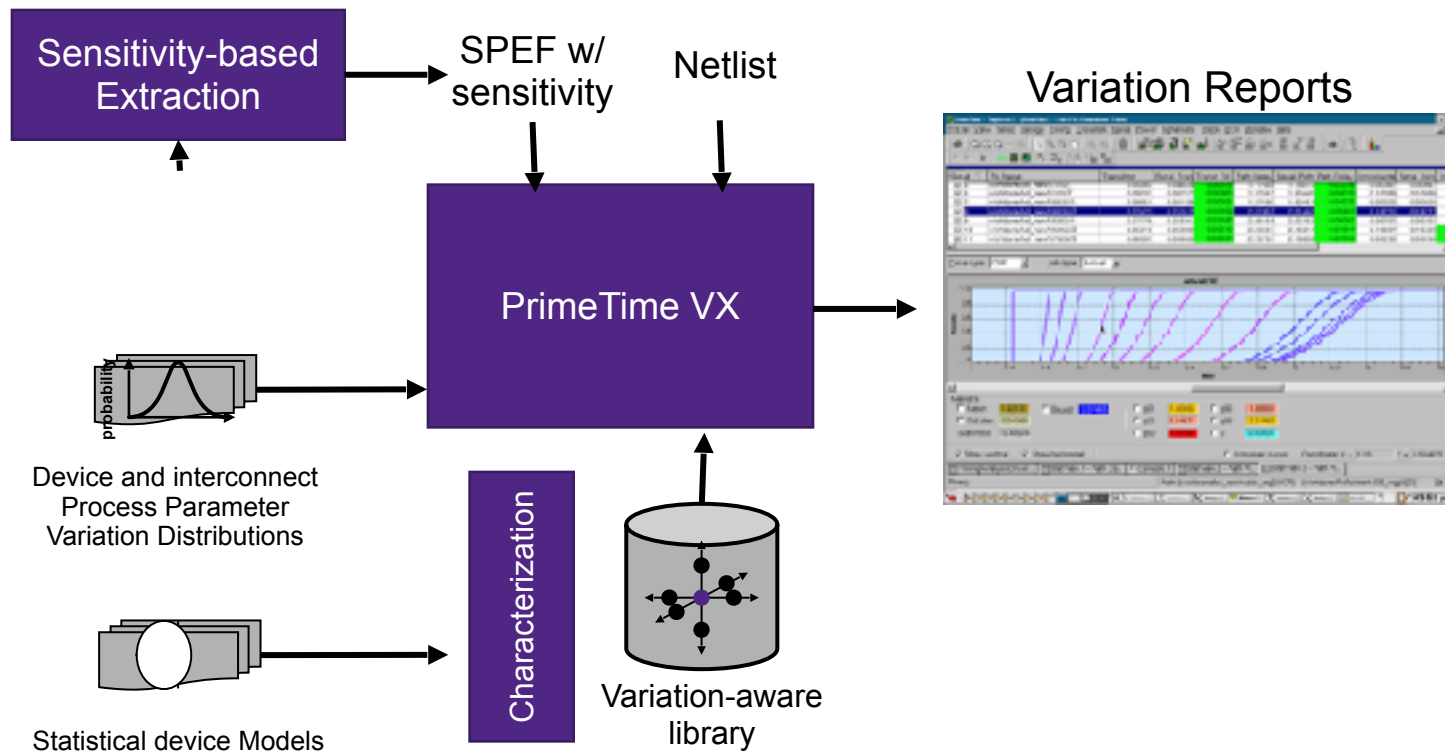
Additional Data for SSTA

Library with Gaussian
distributions if gate
delays

Gate-Netlist



SSTA Tool Inputs and Outputs



Thank You